

Identifying common errors and fixes

Revise 360

Worksheet for the 360° experience: Bug investigation lab

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Bug investigation lab. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Distinguish syntax and logic errors.
- Locate and correct faulty code.
- Use test evidence to verify a fix.

Starter: before you put the headset on

Python print("Ready" is missing a closing bracket. Is the problem a syntax rule or a wrong calculation?

Key terminology

Syntax error

Logic error

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Syntax rules

Turn right

Key facts in your own words:

Example to trace:

```
# Python: faulty  
print("Ready"
```

```
# Correct  
print("Ready")
```

Challenge: What single character fixes this example?

Which is a Python syntax error?

Python `print("Ready")` has...

2 Logic can be wrong

Turn right

Key facts in your own words:

Example to trace:

Need average of a and b

Wrong: $a + b / 2$

Correct: $(a + b) / 2$

For 6 and 10: 8

Challenge: What does the wrong expression give for 6 and 10?

Which expression correctly averages x and y?

For $a = 6$, $b = 10$, $a + b / 2$ gives...

3 Off-by-one errors

Turn around

Key facts in your own words:

Example to trace:

Python list length: 4
Valid indices: 0, 1, 2, 3
Wrong: range(5)
Correct: range(4)

Challenge: Why is index 4 invalid for this list?

Python range(1, 4) repeats how many times?

Last valid zero-based index in a five-item list?

4 Faulty initialisation

Turn around

Key facts in your own words:

Example to trace:

Wrong inside loop:
 total = 0
 total = total + value
Only last value survives

Challenge: Where should total = 0 be moved?

A counter starts at 1 but should count from zero. What type of fault can this cause?

Resetting total in each iteration causes...

5 Trace the faulty line

Turn left

Key facts in your own words:

Example to trace:

Rule: age at least 12

Fault: if age > 12 then

Input: 12

Actual: rejected

Challenge: Give the corrected line and a reason.

Match the fault to its correction.

Match each item to a description from the list below.

Update the counter in the loop / Use >= for at least / Add : after the condition

Item	Matching description
Missing Python colon	
Exclusive threshold	
Counter never changes	

A missing Python colon is corrected by...

6 Verify with evidence

Turn left

Key facts in your own words:

Example to trace:

Fix age ≥ 12

11 -> reject

12 -> accept

13 -> accept

Challenge: Why check 11 as well as 12?

Order the debugging steps.

Put these steps in order: Retest and compare outputs / Apply a targeted correction / Locate and explain the fault / Reproduce the incorrect behaviour

A passing test proves...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Classify the errors.

Categories: Syntax / Logic

Item	Category
Python if $x > 0$ with no colon	
Average calculated as $a + b / 2$	
Missing closing quote	
Discount applied below the wrong threshold	

The corrected average works for 6 and 10. What should you conclude?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 22

Key question

Why does a program running without an error message not prove it is correct?

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.....

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Review and improve

Use review mode. Explain one correction and why the correct answer fits. In Python, test the faulty and corrected average with 6 and 10, then 4 and 4. Expected corrected outputs are 8 and 4.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) State the difference between syntax and logic errors. [2]

b) Correct $\text{average} = a + b / 2$ and explain why the change is needed. [3]

c) A four-item Python list is accessed with `range(5)`. Explain and correct the issue. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Distinguish syntax and logic errors.			
Locate and correct faulty code.			
Use test evidence to verify a fix.			